SOFTWARE DEVELOPMENT TRAINING • CAPSTONE PROJECT

TradeFlow

Intelligent Ethio-Djibouti Trade Corridor Freight Logistics, Telematics & Digital Escrow Platform

Project Name:	TradeFlow Logistics Platform
Version:	v1.1.6 (Production Release)
Document Type:	Final Project Documentation & Report
Submission Date:	November 2026
Repository:	github.com/beki255/trade-flow
API Swagger Docs:	https://tradeflow-api-7abs.onrender.com/api/docs
Android APK:	V1.1.6 PRODUCTION READY

PROJECT GROUP MEMBERS (CAPSTONE TEAM) — 3 Members

STUDENT FULL NAME	STUDENT ID	STATUS
BEREKET SAHLEMARIAM	CTC-1041-26	Group Member
DAGNAW MEKASHA	CTC-1339-26	Group Member
HAMELMAL BELAYNEH	CTC-1116-26	Group Member

🔗 Project Deliverables & Live Links

> **Official Submission Notice:** All project code, server infrastructure, and mobile APK builds are live and fully verified. Below are the primary access URLs required for grading:

Deliverable Name	Environment / Type	Direct URL / Link
GitHub Repository	Source Code (Monorepo)	https://github.com/beki255/trade-flow
Live Web Application	Production Web Portal	https://trade-flow-ukgu.vercel.app
Backend REST API (Swagger Docs)	Cloud Microservice	https://tradeflow-api-7abs.onrender.com/api/docs
API Health Check	Operational Status Check	https://tradeflow-api-7abs.onrender.com/api/health
Mobile Android APK	Installable Standalone APK	Download TradeFlow v1.1.6 APK
EAS Cloud Build Logs	Expo CI/CD Build #5dbf3408	View EAS Build Logs

📱 Instant Mobile APK Camera QR Scan

Scan this QR code with any smartphone camera to instantly download and install the standalone **TradeFlow Android App (v1.1.6)**. No Expo installation required.

Direct APK Download Link →

👥 Capstone Project Group Members & Roles

Student Full Name	Student ID	Academic Role	Core Project Contributions
BEREKET SAHLEMARIAM	`CTC-1041-26`	Group Member	Full-Stack Engineering, Mobile Telematics, Escrow Smart Contract Logic & Cloud Deployment
DAGNAW MEKASHA	`CTC-1339-26`	Group Member	System Architecture, Database Schema Design, REST API Engineering & Testing
HAMELMAL BELAYNEH	`CTC-1116-26`	Group Member	UI/UX Design System, Multi-Language Localization (Amharic & Afaan Oromoo), Documentation

📑 Executive Summary

TradeFlow is a production-ready, cross-platform logistics orchestration, real-time corridor telematics, and digital smart escrow platform purpose-built for the critical **Ethio-Djibouti Trade Corridor**—the 900+ km strategic lifeline carrying over 95% of Ethiopia's import-export commerce.

Historically, this vital trade corridor has suffered from fragmented manual brokerage, exorbitant middleman commission markups (15–25%), zero real-time in-transit visibility, severe cross-border customs bottlenecks at Galafi, payment defaults, and an alarming 42% empty backhaul rate.

TradeFlow resolves these systematic bottlenecks through a fully unified digital ecosystem consisting of:

- **Automated Freight Marketplace:** Eliminates physical brokers, enabling direct carrier-shipper engagement.
- **Digital Smart Escrow:** 100% upfront cargo payment locking with automated release upon verified electronic proof-of-delivery (e-POD).
- **Corridor Telematics & Geofencing:** Continuous GPS positioning and automated milestone check-ins across all 11 corridor transit hubs.
- **Single-Window Customs Clearance:** Digital electronic dossier management with algorithmic risk classification (Green/Yellow/Red Line) and cryptographic QR transit passes.
- **Trilingual Accessibility:** Complete localization across **English, Amharic, and Afaan Oromoo** with native dark mode support.

1. Project Objectives & Scope

1.1 Problem Statement

Over 95% of Ethiopia's international trade flows through the Port of Doraleh in Djibouti to dry port container depots in Modjo and Kaliti. Despite its economic significance, the corridor faces deep operational inefficiencies:

- **Extreme Trust Deficit:** Shippers face persistent risks of cargo theft and demurrage penalties; transporters face delayed payments spanning 30 to 90 days.
- **Blackout In-Transit Tracking:** Once a truck departs Djibouti, cargo owners have no automated means of monitoring location or estimated arrival.
- **Border Customs Congestion:** Manual paper inspection at the Galafi border crossing results in 24 to 72-hour transit halts.
- **High Empty Backhaul Rates:** Trucks returning to Djibouti travel empty 42% of the time, inflating inbound freight costs to over $4,000 per container.

1.2 Core Project Objectives

- **Direct Digital Freight Matching:** Connect cargo owners directly with commercial truck fleets to eliminate middleman margins.
- **Milestone-Based Escrow Security:** Secure freight payments in escrow before vehicle dispatch, releasing funds automatically upon verifiable digital Proof-of-Delivery.
- **Live Telematics & Predictive ETA:** Deliver real-time GPS telemetry and automated checkpoint geofencing across all 11 critical corridor hubs.

- **Digital Single-Window Customs:** Deploy dynamic QR transit passes to reduce border gate inspection time from hours to seconds.
- **Ensure Nationwide Inclusion:** Deliver full trilingual localization in English, Amharic, and Afaan Oromoo with adaptive Dark/Light theme ergonomics.

2. Requirements Specification

2.1 User Roles & Permissions Matrix

Role Identifier	User Persona	Core Responsibilities & System Capabilities
`SHIPPER`	Commercial Importer / Exporter	Publishes freight tenders, funds milestone escrow via Telebirr/CBE, monitors live corridor telemetry, inspects and approves e-PODs.
`TRANSPORTER`	Fleet Owner / Logistics Enterprise	Registers fleet trucks and drivers, submits competitive freight bids, tracks vehicle earnings, and receives automated escrow payouts.
`DRIVER`	Heavy Commercial Truck Driver	Accesses assigned trips on mobile app, performs GPS geofence check-ins, broadcasts corridor incidents, captures e-POD signatures & photos.
`CUSTOMS_OFFICER`	Ethiopian Customs Commission (ECC)	Inspects digital dossiers, classifies risk lines (Green/Yellow/Red), issues digital transit passes, scans driver QR codes at gates.
`ADMIN`	TradeFlow System Operator	Audits TIN and trade license documents, monitors corridor analytics, and oversees dispute resolution.

2.2 Functional Requirements Implementation Summary

- **FR-01 (Authentication & RBAC):** Multi-tenant authentication via JWT tokens, bcrypt credential hashing, and strict role guards.
- **FR-02 (Freight Marketplace):** Dynamic tender listings with cargo specification, target rate, tonnage, and instant bid matching.
- **FR-03 (Smart Escrow Protocol):** Automated funds locking upon contract acceptance; 95% payout release to transporter wallet upon e-POD approval with 5% platform commission retention.
- **FR-04 (Driver Telematics & Geofencing):** Native GPS coordinate collection, Haversine proximity detection (unlocking check-ins within 15 km), and offline queue buffering.
- **FR-05 (Single-Window Customs):** 4-document dossier verification (Declaration, Invoice, Packing List, Insurance), Green Line classification, and offline-capable dynamic QR pass.
- **FR-06 (Electronic Proof of Delivery):** Recipient capacitive signature capture, cargo condition photography, and cryptographic timestamping.
- **FR-07 (Localization & Dark Mode):** Dynamic i18n switching (EN / AM / OM) and synchronized dark mode across all interfaces.

3. Technologies Used & Architecture Justification

Layer	Technology	Justification & Architectural Rationale
Web Frontend	Next.js 14, React 18, Tailwind CSS, Lucide	Server-side rendering, fast first contentful paint, responsive design for logistics dispatchers.
Mobile App	React Native, Expo SDK 52, Expo Location	Cross-platform native performance, low battery GPS background tracking, offline camera & canvas signature.
Backend API	Node.js, Express, TypeScript	Non-blocking asynchronous I/O ideal for concurrent GPS telemetry streams and marketplace events.
Database & ORM	PostgreSQL, Prisma ORM	ACID compliance for escrow financial transactions, type-safe schema migrations, relational integrity.
Real-Time Sync	Socket.IO WebSockets	Sub-second live telemetry distribution between mobile drivers and web dispatch tracking maps.
Cloud Infrastructure	Render (API), Vercel (Web), Neon (Postgres)	Global CDN distribution, automated CI/CD deployments, auto-scaling serverless database.

4. System Design & Database Architecture

The TradeFlow system is organized around normalized relational models ensuring high transaction integrity:

- `User`: Multi-tenant user accounts with role partitioning, TIN credentials, and localized preferences.
- `Load`: Freight tender listings containing pickup, delivery, cargo type, tonnage, and target price in ETB.
- `Bid`: Captures transporter bids, proposed pricing in ETB, and transit timeline commitments.
- `Shipment`: Encapsulates active corridor transit missions, linking the awarded bid, assigned vehicle, driver, and live GPS coordinates.
- `CustomsClearance`: Houses the single-window electronic customs dossier, risk clearance line, certificate number, and digital QR seal.
- `Payment`: Manages the digital smart escrow financial ledger, recording payment method (Telebirr/CBE Birr), locked amounts, and settlement timestamps.
- `IncidentReport`: Records real-time corridor delays and route hazards with GPS stamps, severity levels, and optional photo evidence.
- `Rating`: Bidirectional feedback system enabling shippers and transporters to score each other across verified shipments.

5. Implementation Details & Core Modules

5.1 Haversine Distance & Geofencing Algorithm

```
// Dynamic Proximity Detection (apps/mobile/src/shared/corridor.ts)
const dLat = (liveLat - wpLat) * (Math.PI / 180);
const dLon = (liveLng - wpLng) * (Math.PI / 180);
const a =
  Math.sin(dLat / 2) * Math.sin(dLat / 2) +
  Math.cos(liveLat * (Math.PI / 180)) * Math.cos(wpLat * (Math.PI / 180)) *
  Math.sin(dLon / 2) * Math.sin(dLon / 2);
const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
const distanceKm = Math.round(R * c);

// Check-in unlocks only when driver is within 15 km of designated hub
const isWithinGeofence = distanceKm <= 15;
```

5.2 Milestone Escrow Settlement Logic

When a driver captures the recipient's touch signature and cargo delivery photo, the shipment state transitions to `DELIVERED`. Upon shipper approval, the escrow release method executes atomically:

```
// Atomic Escrow Release Transaction
const platformFee = totalAmount * 0.05; // 5% platform commission
const transporterPayout = totalAmount * 0.95; // 95% net earnings

await prisma.$transaction([
  prisma.payment.update({ where: { id: paymentId }, data: { status: 'RELEASED' } }),
  prisma.transporterWallet.update({ where: { id: transporterId }, data: { balance: { increment: transporterP
  prisma.platformRevenue.create({ data: { amount: platformFee, shipmentId } })
]);
```

6. Features & User Portals

- **Corridor Freight Matching:** Dynamic cargo tenders, automatic route distance calculation, and transporter bidding engine.
- **Smart Escrow Payment Protocol:** 100% upfront payment locking with milestone disbursement upon verified delivery.
- **Live Telematics & Corridor Map:** Interactive Leaflet map displaying active truck markers, speed, heading, and route progress.
- **Driver Mobile Companion App:** GPS geofenced check-ins, offline queue synchronization, customs QR pass, and digital e-POD.
- **Single-Window Customs Clearance:** Digital document upload, ECC risk-tier routing, and dynamic border verification passes.
- **Trilingual Accessibility:** Complete instant language switching for English, Amharic (**አማርኛ**), and Afaan Oromoo.

7. Testing & Quality Assurance

Test Case ID	Module Tested	Scenario & Test Procedure	Observed Result	Status
TC-AUTH-01	Authentication	Multi-tenant registration with role validation	Correct JWT returned with role-based routing.	**PASS**
TC-LOAD-02	Marketplace	Create tender & submit carrier bid	Bid persists in database and notifies shipper.	**PASS**
TC-ESCR-03	Escrow Engine	Shipper locks freight funds via Telebirr	Payment status transitions to `HELD`.	**PASS**
TC-TELE-04	Telematics	Driver GPS ping transmission	Web dashboard updates truck marker position.	**PASS**
TC-GEOF-05	Geofencing	Simulate driver within 15 km of Semera hub	Check-in button enables; check-in logged.	**PASS**
TC-CUST-06	Customs Gateway	ECC Green Line pass generation	Cryptographic QR pass renders on mobile screen.	**PASS**
TC-EPOD-07	Delivery & Payout	Recipient sign & photo capture	95% payout credited to transporter wallet.	**PASS**
TC-I18N-08	Localization	Instant language switch between EN, AM, OM	UI translates instantaneously without page reload.	**PASS**
TC-THME-09	Theme Engine	Toggle Dark / Light theme	High-contrast colors adapt with zero unreadable text.	**PASS**
TC-ANDR-10	Mobile APK	Standalone Android APK installation (v1.1.6)	Installs cleanly on Android 10+ devices.	**PASS**

8. Conclusion & Future Work

The **TradeFlow** platform establishes an innovative technological foundation for cross-border logistics across the Ethio-Djibouti Trade Corridor. By replacing fragmented manual brokerage with an automated freight exchange, securing carrier-shipper transactions via milestone-locked smart escrow, streaming low-latency device telematics, and expediting customs inspection through digital QR passes, the system dramatically reduces transit times, freight costs, and payment delays.

All project requirements set forth by the Software Development Training program have been thoroughly fulfilled. The application has been verified across cloud staging environments and physical Android devices, resulting in production-ready web and mobile deliverables.

Future Enhancements: Future iterations will incorporate direct OBD-II hardware telematics dongles for automated fuel monitoring, AI-driven return trip predictive matching to eliminate empty backhauls, and automated plate recognition (ANPR) cameras at Galafi border checkpoints.

9. Capstone Project Team Credentials & Endorsement

> **Academic Verification:** This software engineering project was designed, developed, and verified collaboratively by the capstone student members listed below:

Student Full Name	Student ID	Academic Role	Individual Capstone Contributions
BEREKET SAHLEMARIAM	`CTC-1041-26`	**GROUP MEMBER**	Full-Stack Engineering, Mobile Telematics, Escrow Smart Contract Logic & Cloud Deployment
DAGNAW MEKASHA	`CTC-1339-26`	**GROUP MEMBER**	System Architecture, Database Schema Design, REST API Engineering & Testing
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